

Computational Design of Proteinous Drug Employing Hidden Markov Model

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1 Introduction

Major histocompatibility complex (MHC) molecules bind antigenic peptides and present them for recognition by the immune system. MHC molecules play an essential role in the initiation and regulation of T cell mediated immune responses. Peptides that bind MHC represent promising targets for the development of vaccines and therapies for autoimmune diseases and cancer. A typical human MHC class II (HLA-DR) binding peptide is 9-35 amino acid long. It has a 9-residue binding core and flanking sequences that protrude from the HLA peptide-binding groove. Prediction methods using weight matrices, artificial neural networks (ANN), or fuzzy neural networks were reported previously [1, 2, 3]. All of these methods require the identification of peptide binding cores for model building. The peptide alignment step is critical because it is computationally difficult and in some cases an incorrect region of a peptide may be assigned as a binding core. In this study, we describe a hidden Markov model (HMM)-based method for prediction of HLA-DR-binding peptides. This method does not require the alignment step for model building.

2 Materials and Methods

2.1 Sequences of antigenic peptides

The training data set consisted of 1435 peptides known to bind or not bind human MHC class II molecule HLA-DRB1*0101 (DR1). These peptides were classified into 4 binding categories (no-, low-, moderate-, or high-affinity). An independent data set comprising 343 peptides was used for model validation.

2.2 Hidden Markov model

The models were built using left-to-right type HMM which are frequently utilized in speech recognition. Peptide sequences of various lengths can be used as HMM input and the information of the 9-mer binding cores regions is not required for the HMM training. We used the successive state splitting (SSS) method for optimization of the structures of HMMs. The SSS method splits a state of the HMM into two states in parallel or in series successively. In this study, the upper limit of the number of states was set to 20. The transition probabilities and symbol generation probabilities have been determined by the Viterbi algorithm.

The method uses both binder models and non-binder models for prediction of peptide binding. The likelihood of a peptide being a binder was determined by assessing the difference of the likelihood of the respective models (Figure 1).

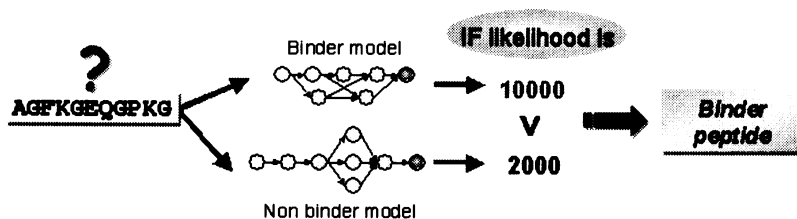


Figure 1: Prediction of peptide binding affinity by the SSS-HMM.

3 Results and Discussions

The predictive ability of the SSS-HMM was compared with the performances of a fully connected HMM (20 nodes) or of an ANN. These models were verified using the ROC analysis. For each model the value of the area under the ROC curve (AROC) is indicated in Figure 2. The SSS-HMM predictions were markedly improved in comparison to the full HMM. This result indicates that the structure of HMM was optimized by the SSS. The ANN predictions were similar to the SSS-HMM predictions for high affinity binders. However, AROC values for SSS-HMM predictions of moderate- and low-affinity peptides were better than the corresponding ANN predictions. Overall, SSS-HMM models achieved higher predictive accuracy than ANN models. These results indicate that amongst the three methods the SSS-HMM is the most effective for predicting MHC class II-binding peptides.

The accurate and reliable models that use only sequence information can help the efficient elucidation of interactions between proteins. We have demonstrated that MHC-binding peptides can be predicted with high accuracy and with minimal pre-processing of peptide data by combining the SSS and HMM methodologies. These tools are important for better understanding of immunity and have applications in vaccine and drug research.

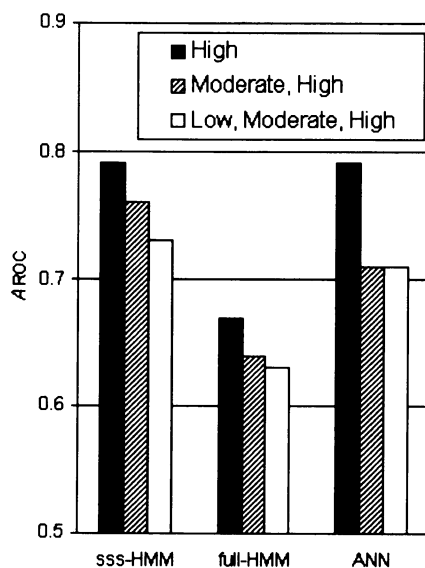


Figure 2: Classification ability of the prediction models.

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